ANDS Project Roadmap

Anomaly-based Network Detection System — task-based build plan

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Workstation structure

```
ands/
■■■ ands.py
■■■ requirements.txt
■■■ core/
■   ■■■ module_base.py
■   ■■■ session.py
■   ■■■ console.py
■■■ modules/
■   ■■■ capture/
■   ■   ■■■ traffic_baseline.py
■   ■■■ detect/
■   ■   ■■■ zscore_anomaly.py
■   ■   ■■■ portscan_detect.py
■   ■■■ report/
■       ■■■ generate_report.py
```

1. Setup

- Fix git init, create the folder structure above
- Confirm venv + pip installs work
- Import Metasploitable2 into VirtualBox, set host-only network, take a clean snapshot
- Run a basic sudo scapy sniff test to confirm the capture chain works
- Push empty skeleton to GitHub

2. Core skeleton

- core/module_base.py — base class (set_option, show_options, run)
- core/session.py — shared state across modules
- core/console.py — minimal use / set / show options / run / back

3. Capture

- modules/capture/traffic_baseline.py — live scapy sniff, BPF filter, feature extraction into a baseline profile
- Pcap save/read support on the capture module

4. Detection

- modules/detect/zscore_anomaly.py — the core detection math

- Unit tests proving z-score catches known synthetic spikes
- modules/detect/portscan_detect.py — second detection angle

5. Validation

- Synthetic traffic generator (normal + attack patterns) — demo safety net
- Full pipeline test against Metasploitable2 — real nmap scan, confirm detection, save pcap as evidence

6. Report & extras

- modules/report/generate_report.py — PDF/HTML report with findings + risk scoring
- Self-test suite (run all modules against known synthetic cases, log false-positive/negative rates)
- Custom module loader + reload command
- JSON/SIEM alert output + uniq command

7. Polish

- Colors ([+]/[-]/[!]/[*]), gobuster-style live-streamed alert output
- Retro CRT boot banner, loading spinner with rotating messages
- Save/delete prompt on run completion, setg, search, Linux command passthrough
- Robustness pass — permission errors, bad interfaces, VPN/interface detection, Ctrl+C cleanup

8. Ship

- pyproject.toml, pip install . working, README
- Writeup: problem, approach, out-of-scope section, security caveats, false-positive rate
- Full demo rehearsal, timed

Work through tasks in order, as many as fit per session. Pick up where you left off next time — no fixed daily quota. Target: finished before September 8.